

Program : Diploma in Communication and Computer Networking	
Course Code: 4328	Course Title: Computer Network Lab
Semester : 4	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Understand how to implement LAN technologies
- Identify network devices
- Identify networking cables and components
- To learn how to configure and understand IP address setting.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	To design and implement LAN technologies	8	Applying
CO2	To identify and use network devices	10	Applying
CO3	To understand network cables and equipment	10	Applying
CO4	To Configure and understand IP address setting	9	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2		1		
CO2	3		2		1		
CO3	3		2		1		
CO4	3		2		1		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To design and implement LAN technologies		
M1.01	To connect and built computers in star topology in a LAN	2	Applying
M1.02	To connect and built computers in ring topology in a LAN	2	Applying
M1.03	To connect and built computers in bus topology in a LAN	2	Applying
M1.04	To connect and built computers in tree topology in a LAN	2	Applying
CO2	To identify and use network devices		
M2.01	To familiarize and identify network devices used in LAN- Hubs, Switches, Routers, Bridges, Repeaters, Gateways, Modems	4	Understanding
M2.01	To connect network devices used in LAN- Hubs, Switches, Routers, Bridges, Repeaters, Gateways, Modems	6	Applying
	Series Test– I	2	
CO3	To understand networking cables and equipment		
M3.01	To study the constructional details of transmission media- co-axial cables, twisted pair cables, optical fiber cable	4	Understanding
M3.02	To feel and know the networking equipment (repeater, hub, bridge, switch, router, crimpers, UTP, Fiber cable, connectors, patch panel, cable managers, racks, CAT6 straight and crossover wiring standards, LAN meter/tester, RJ-45)	6	Applying
CO4	To Configure and understand IP address setting		
M4.01	Configure static and dynamic IP address	4	Applying
M4.02	Run basic TCP/IP utilities and network commands: ipconfig, ping, netstat, pathping, route	5	Applying

	Series Test –II	2	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
Minimum of 6 experiments (excluding open ended experiments/micro projects) shall be performed.
Compulsory for CIA
The CIA shall be arranged in fourth semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
R1	Cisco Lan Switching Fundamentals 2nd Edition by David Barnes, Basir Sakandar
R2	Networking Self-Teaching Guide: OSI, TCP/IP, LANs, MANs, WANs, Implementation, Management, and Maintenance 1st Edition, James Edwards, Richard Bramante
R3	Cabling Part 1: LAN Networks and Cabling Systems, 5th Edition Illustrated Edition by Andrew Oliviero
R4	IP Addressing: IPv4 Addressing Configuration Guide, Cisco IOS XE Release 3S
R5	IP Address Management: Principles and Practice, Timothy Rooney

Online Resources

SI No	Website Link
1	https://onl.kmi.open.ac.uk/
2	https://www.cisco.com/c/en/us/products/cloud-systems-management/modeling-labs/index.html
3	http://vlabs.iitb.ac.in/vlabs-dev/labs_local/computer-networks/labs/explist.php